

4.4 Data Organization and Representation

ٹیبل درست عدد دکھاتی ہے، تعدد کی جدول بتاتی ہے کتنے طلبہ کس رینج میں ہیں، گراف رجحان دکھاتا ہے۔

CS XII

Chapter 4: Data and Analysis

Lecture 06 of 8

4.4 Data Organization and Representation

LEARN IT

Organised data is easier to read. Three everyday forms:

- **Table** — exact values side by side (Ali 85 Math, 80 Science).
- **Frequency distribution** — how many items fall in each range (bin).
- **Graph** — a picture of a trend or comparison (next topics).

Range of marks	Frequency (students)
50–59	5
60–69	7
70–79	9
80–89	4
90–100	2

Most students scored 70–79. That pattern is hard to see in a raw list of 27 marks, and obvious in this table. Matplotlib (`import matplotlib.pyplot as plt`) turns tables into graphs: `import` → `figure` → `plot` → `labels` → `title` → `show()`.

اردو میں سمجھیں

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CHECK YOURSELF

1. How many students scored 80 or above in the table?
2. Name the library that draws graphs in Python.

Answers: 1. $4 + 2 = 6$. 2. Matplotlib (pyplot).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Tabular form is for:

- a. Exact values in rows/columns
- b. Only a pretty poster
- c. Only NAND
- d. Only sprints

MCQ 2. A frequency table shows:

- a. How many in each bin/range
- b. The ER diagram
- c. The OSI stack
- d. The Figma layers

MCQ 3. Most students in the notes' table scored:

- a. 70–79
- b. 90–100
- c. 0–10
- d. Exactly 100

MCQ 4. Graphical representation is for:

- a. Seeing patterns fast
- b. Storing PKs
- c. Compiling C
- d. Routing

MCQ 5. Matplotlib import line:

- a. `import matplotlib.pyplot as plt`
- b. `import pie`
- c. `import osi`
- d. `import nand`

MCQ 6. `plt.show()` :

- a. Renders the figure
- b. Commits SQL
- c. Closes sqlite
- d. Starts Figma

Extra short questions

1. Three organisation methods in the notes.
2. From the frequency table, how many scored 80+?
3. Steps of a matplotlib figure.
4. Why frequencies for large n?

Extra long question

1. Build a frequency table for 12 marks you invent, then say which graph you would draw and why. Include axis titles.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Table, frequency distribution, graph.
2. $4+2=6$.
3. import, figure, plot, labels, title, show.
4. A list of 2000 marks is unreadable; bins are not.

Long

1. Bins of 10. Most-filled bin noted. Histogram (numeric distribution) or column if you treat bands as categories — say which and why bars touch or not.

Next lecture: 4.5 Data Visualization & Types of Graphs